ACOUSTIC SIDE-CHANNEL · DSP 01204496 · DETAILED EDITION

Acoustic Side-Channel Attack on the Thai Keyboard

From the **physics of a struck plate** to a working classifier — every signal-processing step explained, measured honestly, and extended to the Thai Kedmanee layout.

Defensive security research — how well the attack works, and what defends against it

Final project · Computer Engineering, Year 3 · October 2026

THREAT MODEL

What does the attacker actually need?

Has

A single microphone near the keyboard — a phone on the desk is enough. Recordings of the victim typing.

Does NOT need

Any software on the victim's machine, network access, or physical contact. The leak travels through the air as sound.

At stake

Passwords, messages, private text — anything typed within earshot.

This study is **defensive**: we build the attack only to measure how feasible it is and what breaks it.

PRINCIPLE — THE PHYSICS

A keyboard is a plate; each key strikes it at a different point

A struck plate vibrates in a set of **modes**. Striking at a different position excites those modes in different proportions, so each key radiates a different mix of frequencies — a timbre.

A key's sound is the plate's **fixed vibration modes** — their frequencies and how fast they fade — each weighted by how strongly that key's **position** excites it.

Mode frequencies & decay times — fixed by the hardware.

Excitation weights — set by the key's **position** on the plate.

The character printed on the key never enters — only its location does.

So it is a **signal** problem, not a language problem — a classifier trained on English keys transfers to Thai by relabelling.

METHOD — OVERVIEW

Five stages, microphone → predicted key

1
Record
48 kHz + keylog, one clock

→

2
Detect
envelope + onsets (FIR)

→

3
Window
64 ms per press

→

4
Feature
log-mel spectrogram

→

5
Classify
SVM + k-fold CV

Stages 1–4 are pure DSP (syllabus weeks 2, 8, 3, 5, 7). Stage 5 is deliberately a simple classifier, so the contribution stays in the signal processing.

The next slides open each stage and explain what it does, and why.

STAGE 1 — RECORDING WITH FREE LABELS

Sample the air, and timestamp every key

Audio is sampled at **48 kHz**. By the Nyquist limit that captures frequencies up to **24 kHz** — far above the key-click energy, most of which is below 10 kHz.

Sampling just means measuring the microphone 48,000 times a second and storing the stream of numbers.

A keylogger records each key with its timestamp on the **same clock**. Recording audio and keys together labels every press for free — no manual annotation.

The SNR caveat

Sampling is easy; the real limit is **signal-to-noise ratio**. A close mic hears a crisp click; a far one hears room noise. We quantify this later.

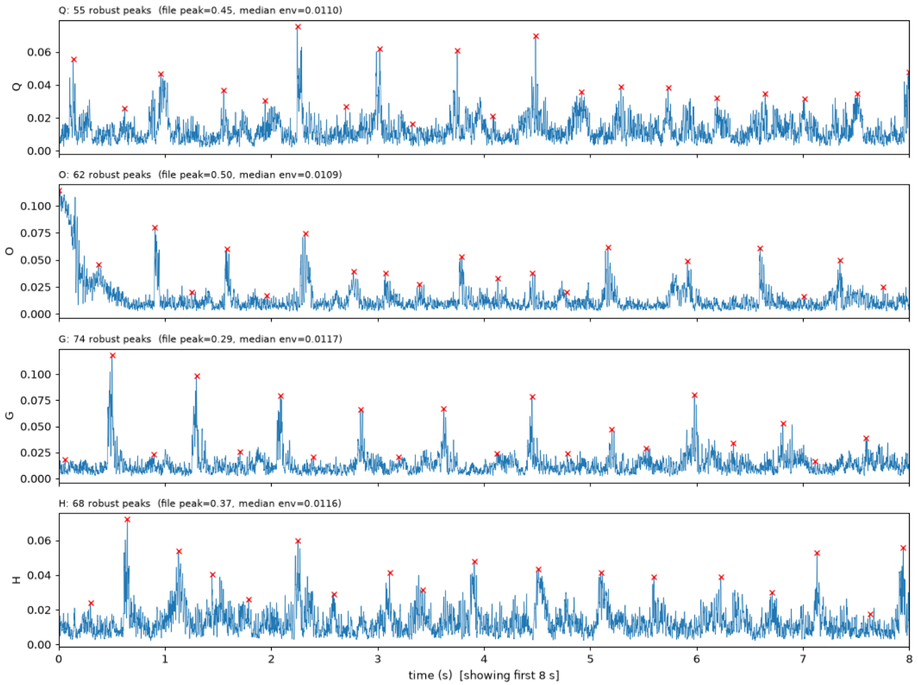
STAGE 2A — ONSETS VIA AN FIR ENVELOPE

Rectify, then smooth with a moving-average FIR filter

Energy envelope = **rectify** the waveform (fold the negatives up), then **smooth** it with a flat 5 ms moving-average filter.

An **FIR filter** replaces each sample with a weighted sum of its neighbours — convolution with a short kernel. The flat kernel is the simplest low-pass; it turns the jittery waveform into a slow envelope that bumps up on every strike.

Rectify first, or the + and − swings average to zero. Red × = detected onsets.



A peak counts only if it stands out locally

Keep an envelope peak only if it is **locally prominent** (stands out from its surroundings) **and** at least 0.3 s from the previous peak.

Prominence, not height

A single clipped spike would make a "fraction-of-max" threshold miss every real press. Prominence is **local**, so one loud spike can't hide the rest.

Refractory spacing

One press has a loud strike and a quieter release. Requiring ≥ 0.3 s between peaks stops one press being counted twice.

This single robustness fix took label quality from a few percent to ≈ 99% of presses matched.

STAGE 2C — ALIGNMENT

Cross-correlation lines the keylog up to the audio

The audio clock and the keylog clock start a little apart. We find the time shift that makes keystroke times land on detected onsets.

Cross-correlation: slide the keystroke times across the audio onsets; measure the overlap at every shift. The shift with the most matches is the true clock offset.

Computed fast through the FFT (the convolution theorem), so it ties back to Stage 3.

Each logged key then snaps to its **nearest unused** onset within a 0.12 s tolerance — one sound, one label.

Taper each frame so the FFT doesn't invent frequencies

Hann window: multiply the frame by a smooth bell that tapers to zero at both ends, so its start and end meet smoothly.

The FFT assumes each frame **repeats forever**. If the two ends don't match, that wrap-around is a sudden jump → fake broadband energy, called **spectral leakage**.

Tapering the ends removes the jump, so energy stays at the true frequencies.

Our frames

Frame = 768 samples (16 ms)
Hop = 384 samples (8 ms, 50% overlap)
≈ **7 frames** per 64 ms press

STAGE 3B — SPECTRUM & PERIODOGRAM

From the windowed frame to power at each frequency

The **FFT** gives amplitude and phase at every frequency. The **periodogram** is the magnitude squared — power at each frequency, with phase thrown away.

FFT

Complex — carries amplitude and phase at each frequency bin.

Periodogram

Magnitude squared — keeps **power**, the part that fingerprints a key's timbre.

Phase shifts with exactly when the strike lands in the frame — not useful for identity — so discarding it makes the feature more stable.

STAGE 3C — MEL FILTERBANK

Compress hundreds of bins into 40 perceptual bands

The **mel scale** spaces frequency the way we hear — fine detail low, coarser high.

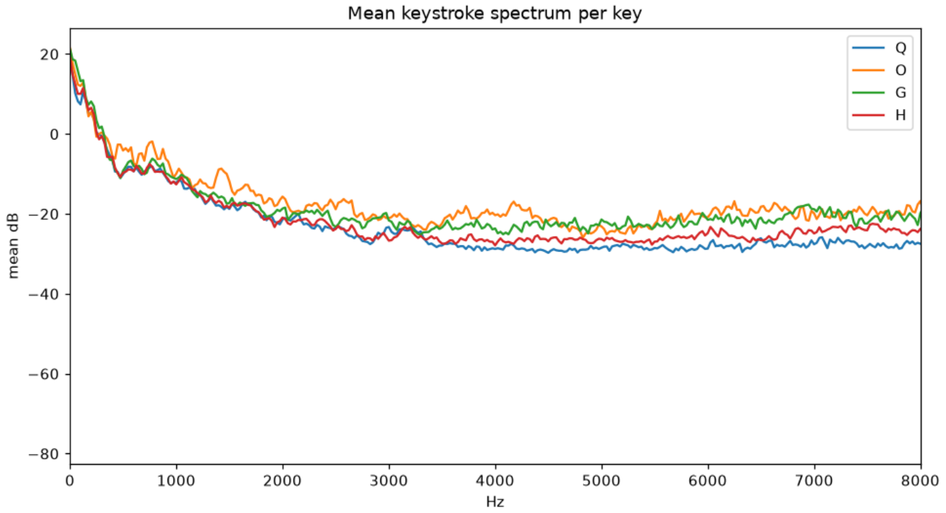
40 triangular filters sum the power spectrum into 40 bands; a **log** then puts loud and soft presses on one scale.

So a frame of hundreds of frequency bins becomes just 40 robust numbers.

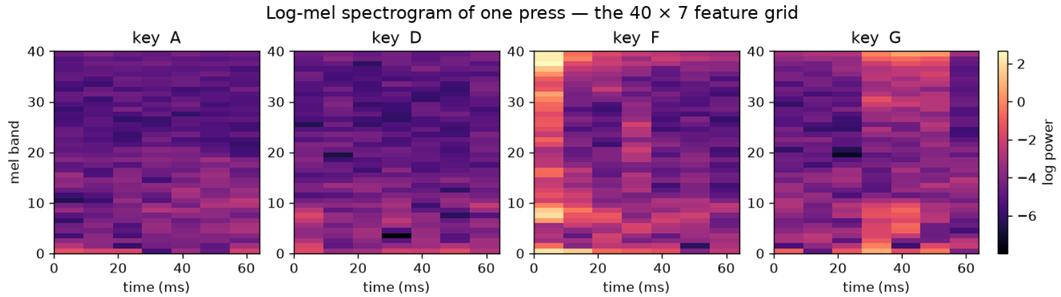
Why it helps

Fewer features → less overfitting · neighbouring bins averaged → less noise · the discriminative detail is kept.

One frame loses the story; seven keep it



Averaged → keys overlap → single spectrum 33.7%



Over time → keys differ → spectrogram 51.8%

The identity lives in the **time evolution** — attack then decay. Each press becomes a 40×7 grid, flattened into a **280-number fingerprint**.

STAGE 4 — THE CLASSIFIER

A max-margin SVM with an RBF kernel

Every press is a point in 280-dimensional space. The **SVM** carves that space into one region per key with the widest possible margins; a new press is labelled by the region it lands in.

The **RBF kernel** lets those boundaries curve — it scores two presses as similar when their fingerprints are close. Features are z-scored first.

Why SVM, not a neural net

Many features, few samples (~50 presses/key). A margin classifier generalises there; a neural net needs far more data. Measured on the next-to-last slide.

MEASURING HONESTLY

Stratified 5-fold cross-validation

Accuracy is the **average over the 5 folds**; "chance" is 1 divided by the number of keys — the score a random guess would get.

Never test on training

Split presses into 5 parts; train on 4, test on the held-out 1; rotate and average. "Stratified" keeps each key's share equal in every fold.

Within-session caveat

Train and test come from the same recording. Real-world use is cross-session — far harder (slide 19).

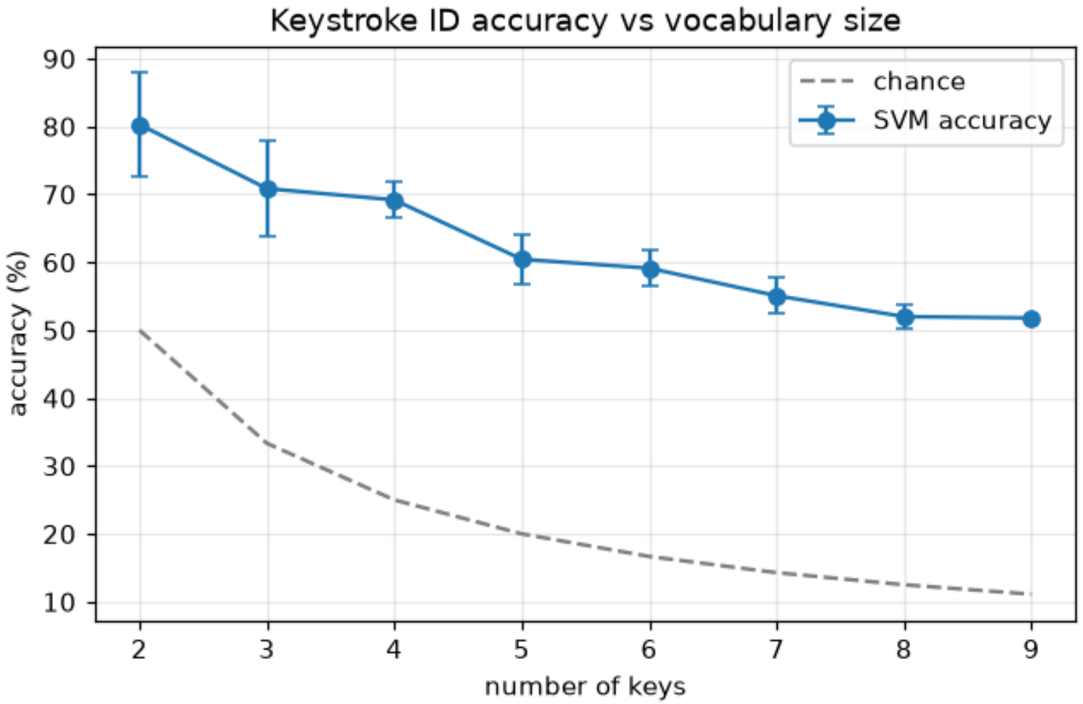
Confusion matrix: rows = true key, columns = prediction. A bright diagonal = correct; the off-diagonal shows which keys get mistaken for which.

RESULTS — ENGLISH

More keys is harder, but always far above chance

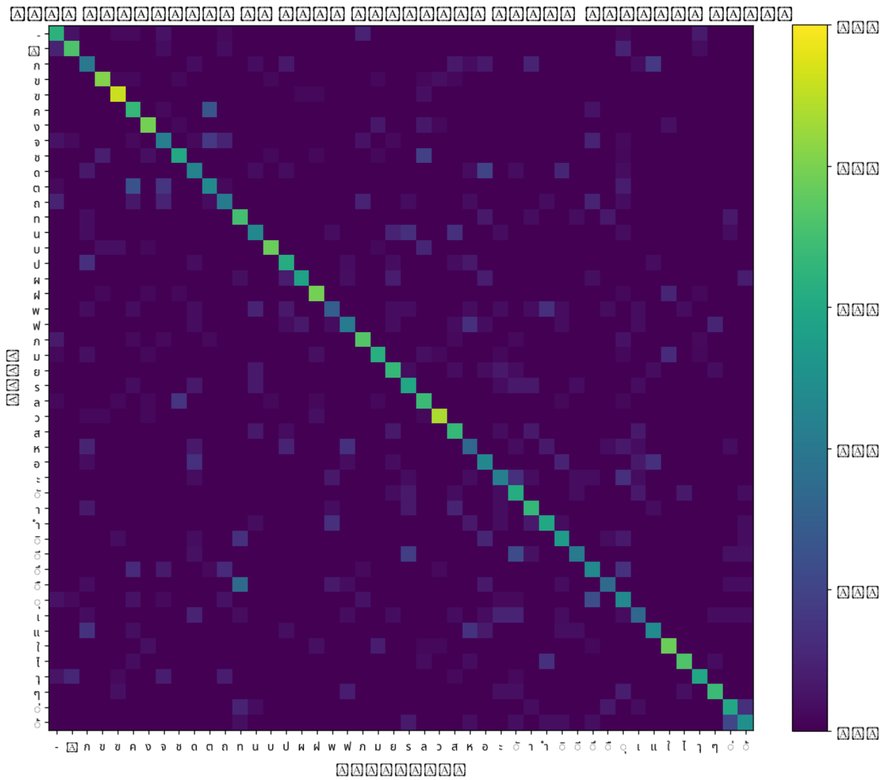
9 keys	**86%**
36 keys (letters+digits)	**52%**

Chance is only 11% and 2.8% — the gap over guessing **widens** as keys grow. Within-session, 5-fold CV.



THAI EXTENSION — THE NOVEL ANGLE

Kedmanee 46 keys: relabel only, same accuracy



Kedmanee 46 keys (full main block)

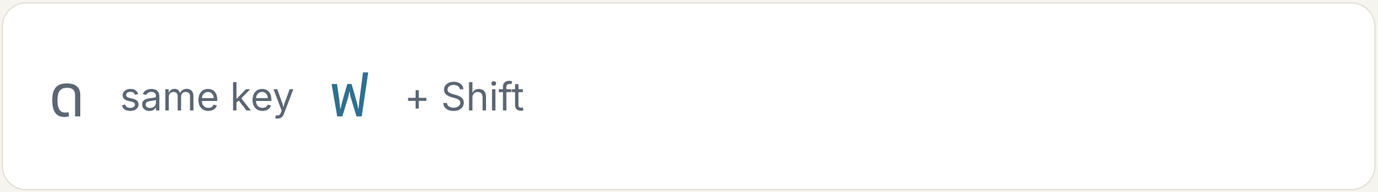
60.5%

Chance 2.2% — ≈ 27× better · 1,626 presses · 5-fold CV

Sound follows the physical key, so the English classifier **is** the Thai classifier — map each key to its Kedmanee character.

SHIFT LAYER — THAI-SPECIFIC

Two characters per key: was Shift held?



A binary question answered from the Shift key's own click, just before the letter. Recorded with normal/Shift order **randomly interleaved** to remove the typing-rhythm confound.

Shift-detection accuracy

74.6%

Chance ~60%

MICROPHONE EXPERIMENT

The ceiling is sound quality, not the algorithm

Same session, two mics. Every algorithm identical — only the microphone changed.

Laptop

35%

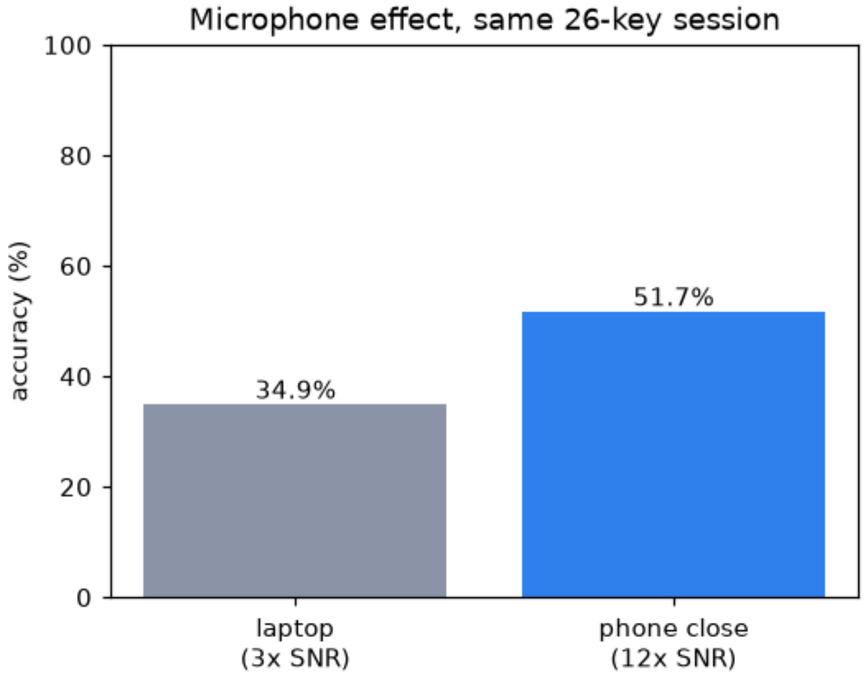
low SNR

Phone near keys

52%

high SNR

+17 points from the microphone alone — this is the gap to the 95% papers.



LIMITS & DEFENSES

Harder in the real world — which is the defense

Same session

95%

Cross-session (real world)

34%

Domain shift

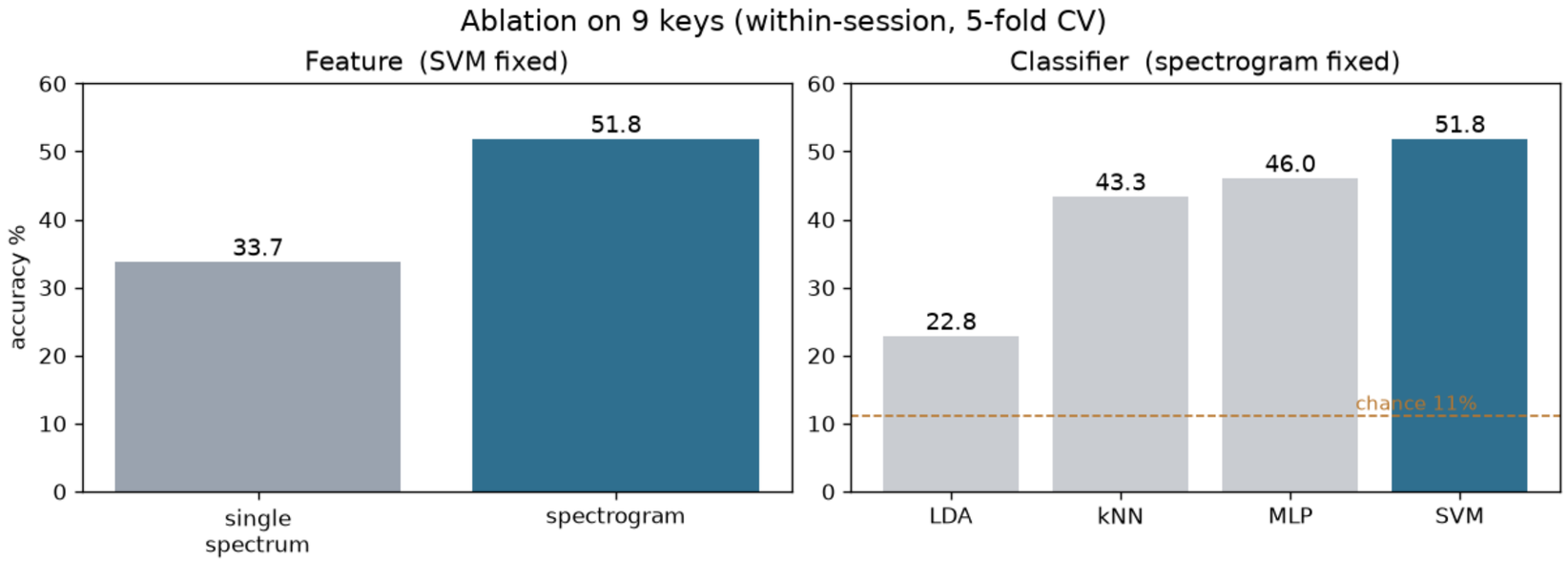
mic moves · force changes · room changes → accuracy drops

Defenses

Type in a noisy environment · quiet / membrane keyboard · the attack is fragile to any change of device or typist.

Evaluated offline on held-out data, as the literature does (Asonov 2004; Harrison 2023) — a live demo is not how real results are reported.

Change one thing at a time; the gap is the story



The **feature** is the biggest lever; a classic SVM beats a neural net when data is scarce.

CONCLUSION & THANKS

In-syllabus DSP is enough — on the Thai keyboard

- The feature (spectrogram) is the heart
- A classic SVM beats a neural net at this data scale
- Sound quality (SNR) is the ceiling

Thank you — questions welcome